

Lab 09 — Well-Architected Review & Cost Controls

Review a small architecture against the Well-Architected pillars: find the exposure, prove the single point of failure, fix it by scaling out, and write up the findings.

Lab environment: Docker Desktop (free local simulation) — runs locally, no AWS account needed

Estimated time: 35–40 minutes

SAA-C03 domain: All four SAA-C03 domains

Overview

Local component	AWS equivalent
lab9-proxy (NGINX, port 8089)	Application Load Balancer
app service (scalable, no fixed name)	EC2 / ECS behind an Auto Scaling group

Step 1 — Get the lab source and start the lab

The repo lives in `~/Downloads/aws-saa`. The paths are absolute, so these commands work from any folder. `git clone` is needed only once: if the repo is already there it prints `already exists`, and the next line still takes you into this lab.

```
cd ~/Downloads
git clone https://github.com/mohanpothula/aws-saa.git
cd ~/Downloads/aws-saa/labs/lab-09-well-architected-review
docker compose up -d
docker compose ps
```

Step 2 — Security review

Task: Find the published ports.

```
docker compose ps
```

Expected result: You identify every externally exposed port, including the one on `app` itself.

Step 3 — Reliability review

Task: Stop the app and observe the impact.

```
docker compose stop app
curl -f http://localhost:8089 || echo 'SERVICE DOWN'
```

Expected result: `SERVICE DOWN` — a single instance is a single point of failure.

Step 4 — Improve reliability

Task: Scale the app to two replicas behind the proxy.

```
docker compose start app
docker compose up -d --scale app=2
```

Expected result: Two `app` containers are running.

Step 5 — Performance review

Task: Run 50 requests.

```
for i in $(seq 1 50); do curl -s http://localhost:8089 >/dev/null; done
echo done
```

Expected result: All requests complete.

Step 6 — Well-Architected findings

Task: Document one Security, Reliability, Performance and Cost improvement.

```
echo 'Security: ... Reliability: ... Performance: ... Cost: ...'
```

Expected result: Four justified findings.

Verification

Check	Expected
Step 2 — Security review	You identify every externally exposed port, including the one on app itself
Step 3 — Reliability review	SERVICE DOWN — a single instance is a single point of failure
Step 4 — Improve reliability	Two app containers are running
Step 5 — Performance review	All requests complete
Step 6 — Well-Architected findings	Four justified findings

Completion check

- [] Every validation result demonstrated to the trainer
- [] You can say which AWS service each local component represents
- [] Your four findings each name a pillar and a concrete change

Cleanup

```
docker compose down -v
```

-v also removes the lab's volumes, so the next run starts from a clean state.

Reference

- Lab source code: <https://github.com/mohanpothula/aws-saa/tree/main/labs/lab-09-well-architected-review>
- Original student guide: [courseware/AWS_SAA_Lab9_Student_Guide.docx](#)
- [AWS Well-Architected Framework](#)
- [Well-Architected Cost Optimization Pillar](#)
- [What is an Application Load Balancer?](#)
- [Amazon ECS Developer Guide](#)
- [Docker Compose](#)
- [NGINX proxy module \(proxy_pass, proxy_cache\)](#)